Decision Mirror – Task 4

Input Specification, Output Specification, and JSON Schema

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1. Input Specification

Purpose: Defines the formal request contract between the application layer and the LLM-based trait extraction component.

Field Name	Type	Required	Description
decision_id	string	Yes	Unique decision identifier
user_id	string	Yes	Unique user identifier
source_type	string	Yes	Source of the input data
decision_text	string	Yes	Free-text decision content
ontology_version	string	Yes	Trait ontology version
timestamp	string	No	Decision creation timestamp
language	string	No	Input language
context	object	No	Additional situational context

Allowed Values:
source_type = decision_log | onboarding_quiz

Input Validation Rules:
- decision_id must not be empty.
- user_id must not be empty.
- decision_text must contain textual content.
- ontology_version must be provided.
- source_type must be either decision_log or onboarding_quiz.
- Unsupported fields must be ignored.

2. Output Specification

Purpose: Defines the machine-readable response returned by the LLM-based trait extraction component.

Field Name	Type	Required	Description
decision_id	string	Yes	Processed decision identifier
extraction_status	string	Yes	Extraction result status
traits	array	Yes	Extracted trait list
model_version	string	Yes	Model version identifier
prompt_version	string	Yes	Prompt version identifier

Trait Object:
trait_id, trait_name, score, confidence, evidence, context_notes

Supported Traits:
T01 Risk Tolerance
T02 Time Horizon
T03 Locus of Priority
T04 Decisiveness Speed
T05 Emotional vs. Analytical Weighting
T06 Self-Consistency

Output Validation Rules:
- extraction_status: success, insufficient_evidence, error.
- score range: 0.0–1.0.
- confidence range: 0.0–1.0.
- evidence must support the assigned score.
- T06 must be null when insufficient history exists.
- malformed outputs must be rejected and logged.

3. JSON Schema


```
{
 "$schema": http://json-schema.org/draft-07/schema#,
 "title": "DecisionMirrorTraitExtraction",
 "type": "object",
 "required": [
  "decision_id",
  "extraction_status",
  "traits",
  "model_version",
  "prompt_version"
 ],
 "properties": {
  "decision_id": {
   "type": "string"
  },
  "extraction_status": {
   "type": "string",
   "enum": [
    "success",
    "insufficient_evidence",
    "error"
   ]
  },
  "traits": {
   "type": "array",
```

```
"items": {
  "type": "object",
  "required": [
    "trait_id",
    "trait_name",
    "score",
    "confidence",
    "evidence"
  ],
  "properties": {
    "trait_id": {
      "type": "string"
    },
    "trait_name": {
      "type": "string"
    },
    "score": {
      "type": "number",
      "minimum": 0.0,
      "maximum": 1.0
    },
    "confidence": {
      "type": "number",
      "minimum": 0.0,
      "maximum": 1.0
    },
```

```
        "evidence": {
          "type": "string"
        },
        "context_notes": {
          "type": "string"
        }
      }
    }
  },
  "model_version": {
    "type": "string"
  },
  "prompt_version": {
    "type": "string"
  }
 }
}
```